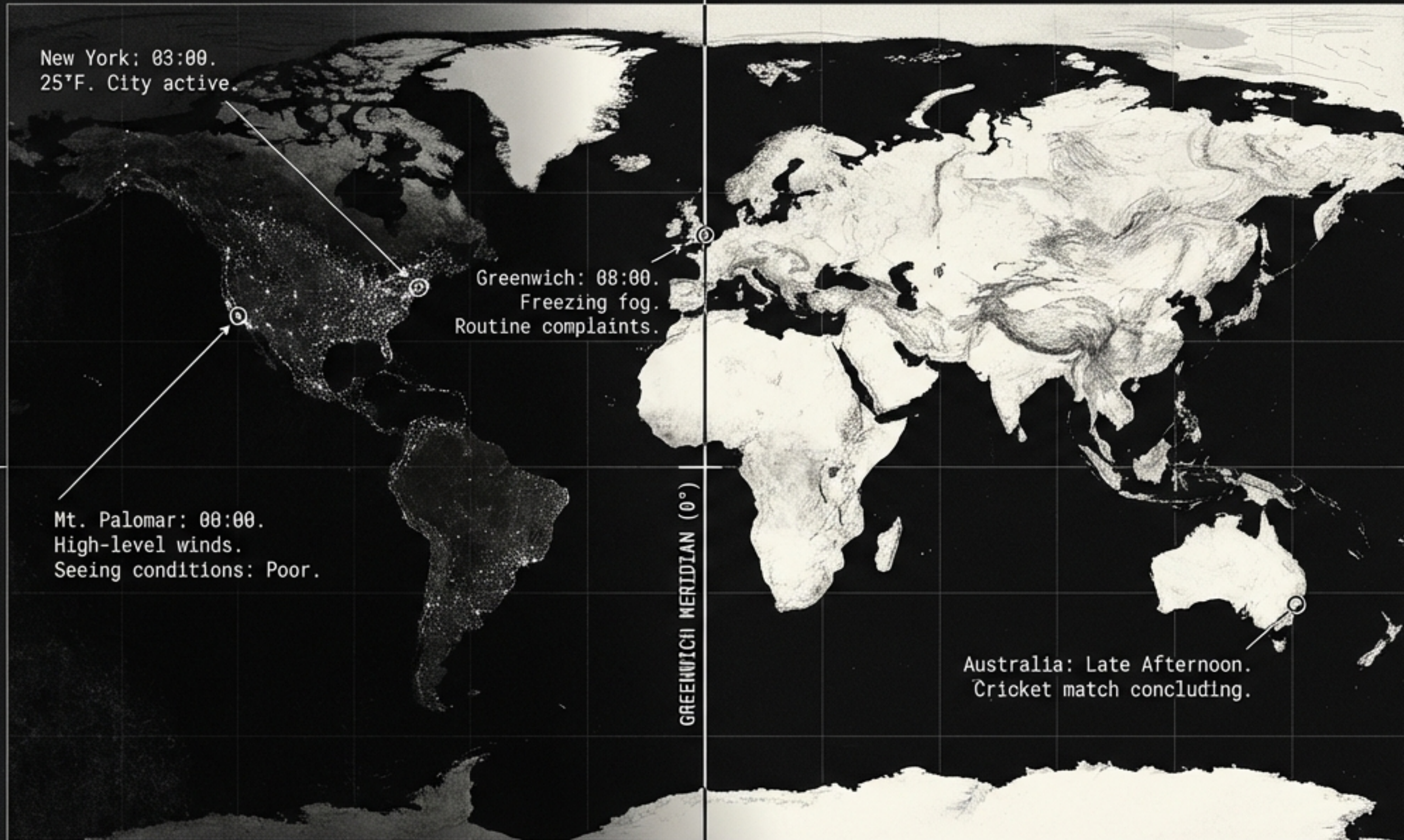


January 7, 1964: A Cold Morning on the Greenwich Meridian



STATUS: NORMALCY

At the Mount Palomar Observatory, 120 miles south of Los Angeles, astronomers Bert Emerson and Paul Rogers break for a midnight meal of soup and sardines. The wind howls against the dome. It is the last hour of the pre-discovery era.

THE ANOMALY ON THE 18-INCH SCHMIDT

OBSERVER: Knut Jensen,
Student (24)
TARGET: Supernovae search,
South of Orion.
METHOD: Blinker Comparator
(Rapid switching
between plates).

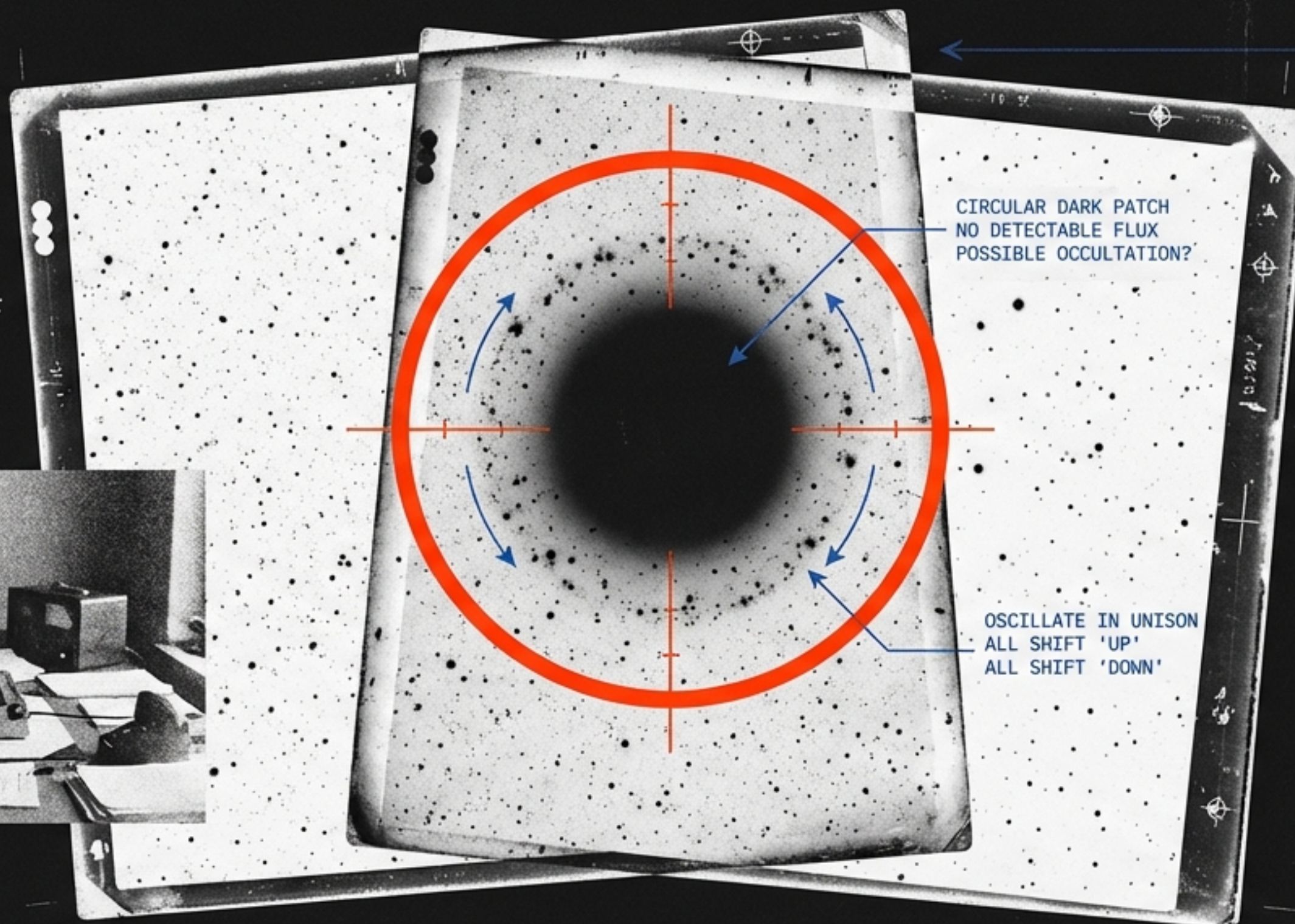
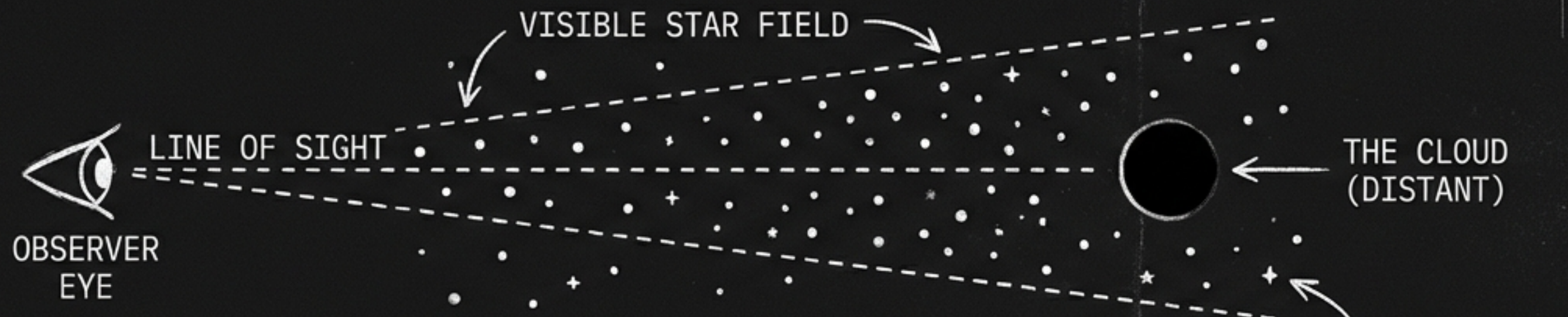


PLATE A: Dec 9, 1963
PLATE B: Jan 7, 1964
DISCOVERY: A circular
dark patch, approx 2.5
degrees diameter.

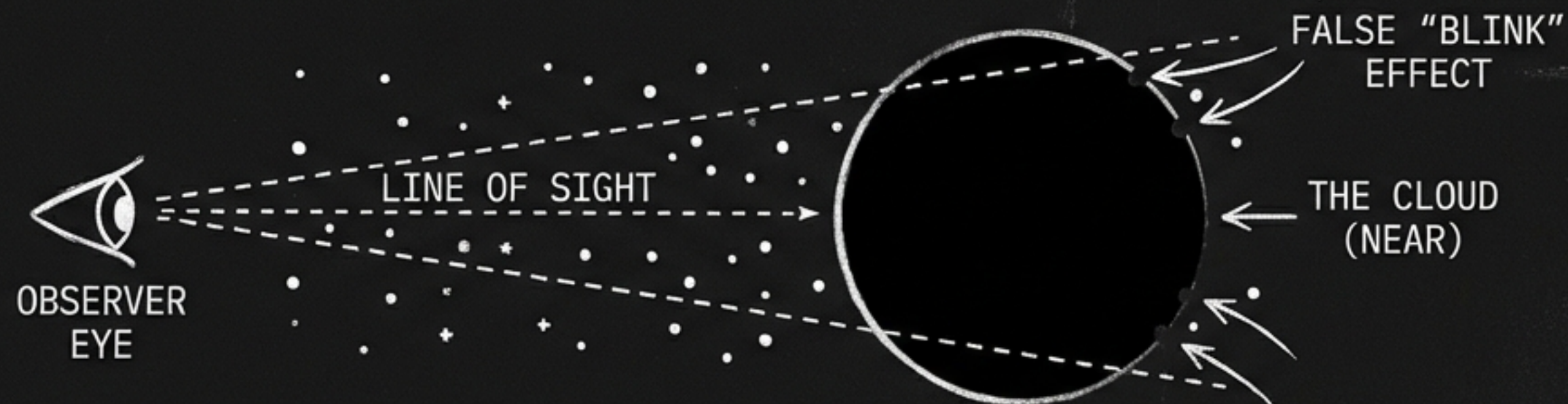
THE IMPOSSIBILITY:
The ring of stars
surrounding the patch
appears to oscillate in
unison. All shift 'up'
together, all shift
'down' together.

IT IS NOT THE STARS THAT ARE MOVING

STATE 1
(DISTANT CLOUD)



STATE 2
(NEAR CLOUD)



ANALYSIS (DR. MARLOWE):

HYPOTHESIS A: Variable stars oscillating in phase.
(REJECTED: Statistically Impossible).

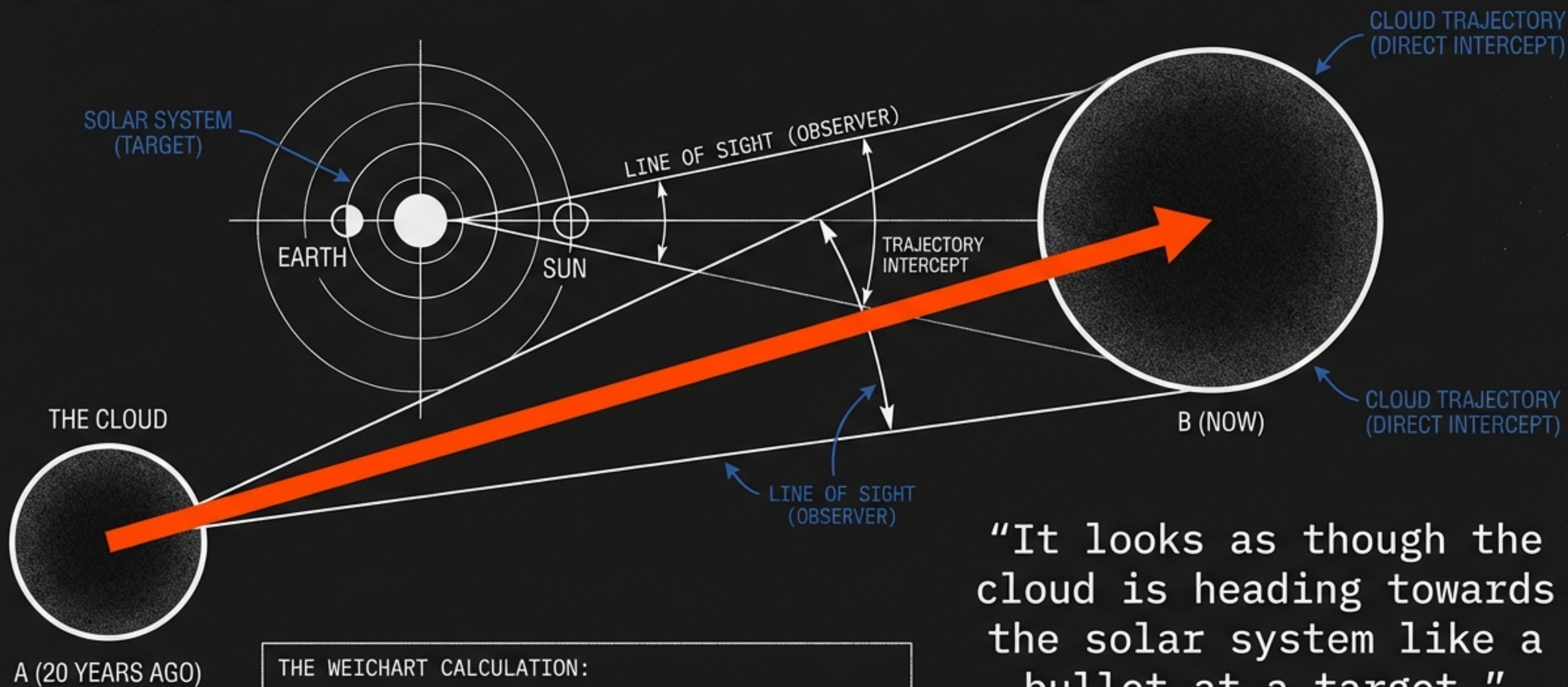
HYPOTHESIS B: A dark object moving towards Earth. (ACCEPTED).

MECHANISM: As the object advances, it eclipses background stars at the edges, creating a false "blink" effect.

CONFIDENTIAL - EYES ONLY

PROJECT: ORION ANOMALY

EVIDENCE OF A BULLET AT THE TARGET



THE WEICHART CALCULATION:

- LATERAL MOTION: < 0.25 degrees in 20 years.
- RADIAL MOTION: Massive increase in angular diameter.
- VELOCITY VECTOR: Almost zero transverse velocity.

"It looks as though the cloud is heading towards the solar system like a bullet at a target."

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PROJECT: ORION ANOMALY

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MEANWHILE IN LONDON: THE B.A.A. MEETING

Royal Astronomical Society Rooms, Burlington House

SPEAKERS: 1. Dr. Oldroyd (Chair);
2. Mr. George Green (Amateur Observer);
3. Professor Chris Kingsley (Cambridge - Skeptic)

DISCREPANCIES IN PLANETARY POSITIONS:

JUPITER: Longitude +1m 29s | Declination -49s

SATURN: Longitude +42s | Declination -17s

KINGSLEY'S REACTION: "Preposterous." Errors of this magnitude are impossible unless the laws of gravity have changed.

THE IMPOSSIBLE ERRORS ARE REAL

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PROJECT: ORION ANOMALY

CONFIRMATION:
Royal Observatory
confirms amateur data.

PLANET	DISCREPANCY IN LONGITUDE	DISCREPANCY IN DECLINATION
JUPITER	+ 1 minute 29 seconds	- 49 seconds
SATURN	+ 42 seconds	- 17 seconds

ADDITIONAL ANOMALIES:
Uranus and Neptune
also out of position.

CORROBORATION:
Heidelberg Observatory reports
identical deviations.

THE HYPOTHESIS: A hitherto unknown body must exist in the vicinity of the solar system... comparable with or even greater than the mass of Jupiter.

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ONE NIGHT ON THE CAMBRIDGE COMPUTER

T Z 0 A 23 Ø 1 U 11 Ø 2 A 2 F 3 U 13 Ø

INPUT: Planetary Discrepancies (Jupiter, Saturn, Uranus).
(#2E5FAC)

PROCESS: Inverse Perturbation Analysis.

COMPUTATIONAL SPEED: 100,000 ten-figure numbers per minute.
(#2E5FAC)

OUTPUT: Mass, Position, Velocity of 'The Intruder'.
(#2E5FAC)

RESULT: A single object
explains ALL deviations.

THE CONVERGENCE OF EVIDENCE

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UK/MATH STREAM

SOURCE: Cambridge Calculation

COORDINATES: RA 5h 46m, Dec $-30^{\circ} 12'$

METHOD: Gravitational Perturbation

Ref: Inverse Perturbation Analysis (IPA)
confirmed deviations.



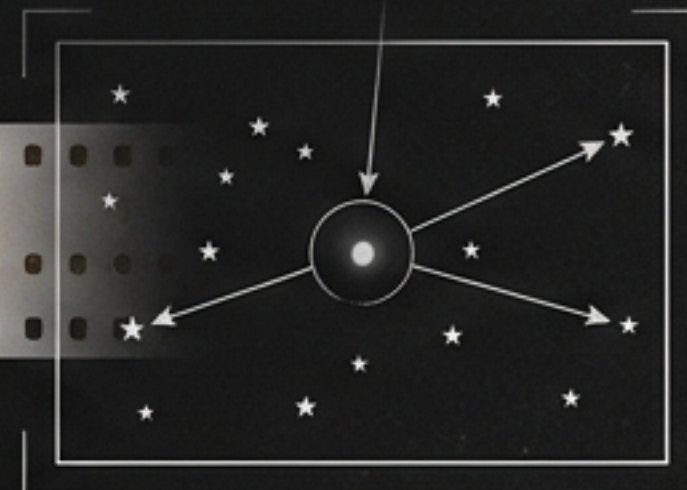
US/VISUAL STREAM

SOURCE: Mt. Palomar Observation

COORDINATES: RA 5h 49m, Dec $-30^{\circ} 16'$

METHOD: Visual Photography

Ref: Palomar Schmidt Telescope Plate #4023A
reveals anomaly.



IMPERATIVE YOU AND ASTRONOMER ROYAL
COME IMMEDIATELY TO PASADENA.

CONCLUSION:

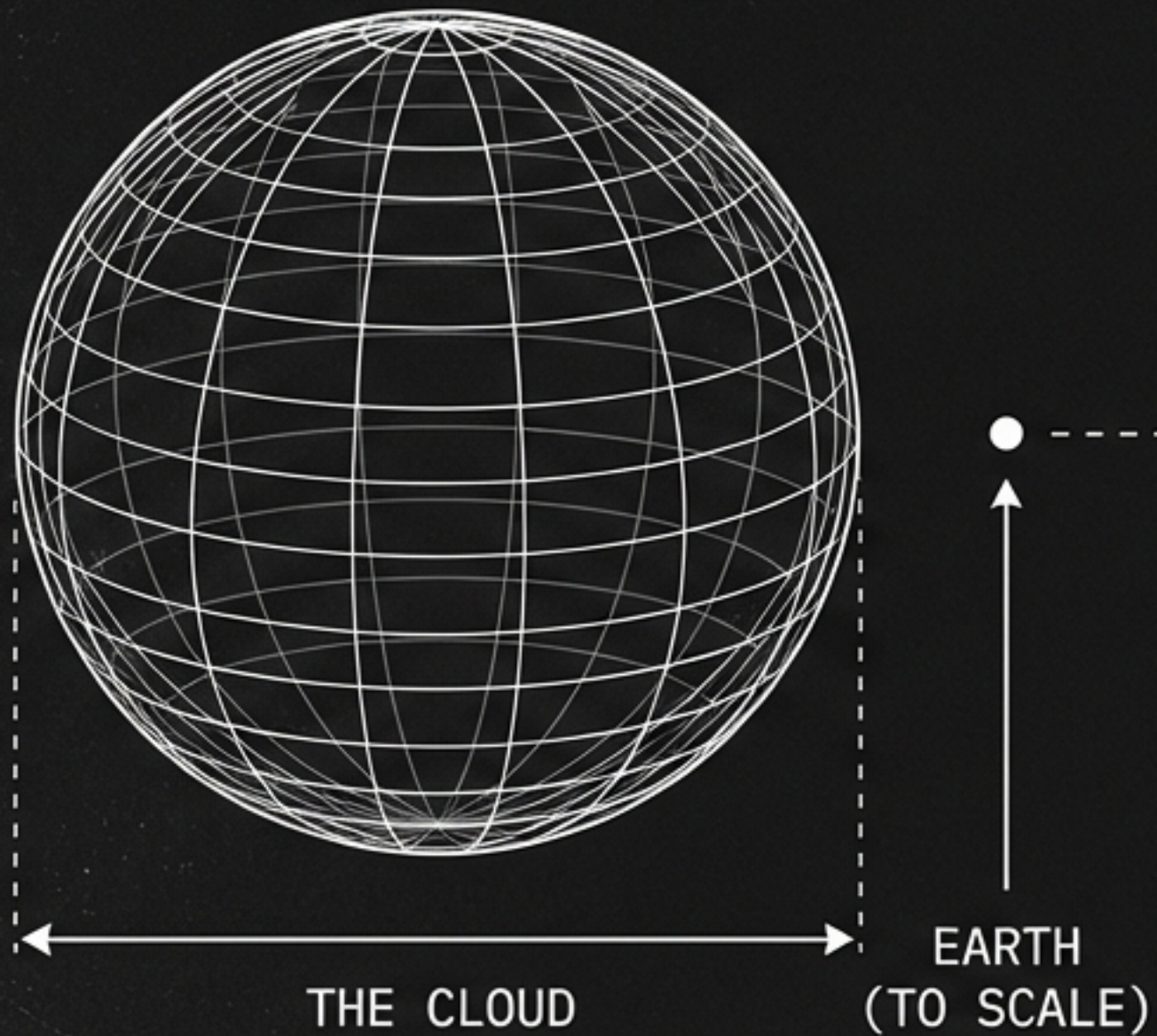
The 'Intruder' (Math) and the 'Black Cloud' (Visual) are the same object.

DISTANCE: 21.3 AU.

VELOCITY: 70 km/sec.

Convergence confirmed by cross-referencing multiple observatories.

PROFILE OF THE INTRUDER



TYPE:	Hydrogen Gas Cloud (Neutral).
DIAMETER:	~1.0 Astronomical Unit (Distance from Sun to Earth).
MASS:	1.3×10^{30} grams (Approx. 2/3 Jupiter).
DENSITY:	1.3×10^{-10} gm/cm ³ .
OPACITY:	Total. (High enough to block sunlight).
CURRENT POSITION:	South of Orion, approaching Solar System.

NOTE: Diffuse for a planet, but extremely dense for a nebula.

ARRIVAL ESTIMATE: AUGUST 1965

CONFIDENTIAL
- EYES ONLY

WEICHART CALCULUS DERIVATION

DERIVATION OF TRANSIT TIME (T):

Write α for present angular diameter (radians), d for linear diameter.

D = distance, V = velocity of approach.

$\alpha = d/D$.

Differentiating with respect to time t : $(d\alpha/dt) = (-d/D^2)(dD/dt)$.

$V = -(dD/dt)$, thus $(d\alpha/dt) = (d/D^2)V$.

Since $D/V = T$, we have $(d\alpha/dt) = (d/DT)$.

Therefore, $T = \alpha (dt/d\alpha)$.

Approximation: $dt/d\alpha \approx \Delta t/\Delta\alpha$. Given $\Delta t = 1$ month, $\Delta\alpha/\alpha = 0.05$.

$\Delta t/\Delta\alpha = 20$.

$T = 20 * 1 \text{ month} = 20 \text{ months}$.

VELOCITY: 70 km/sec → Accelerating
to 80 km/sec (Solar Gravity).

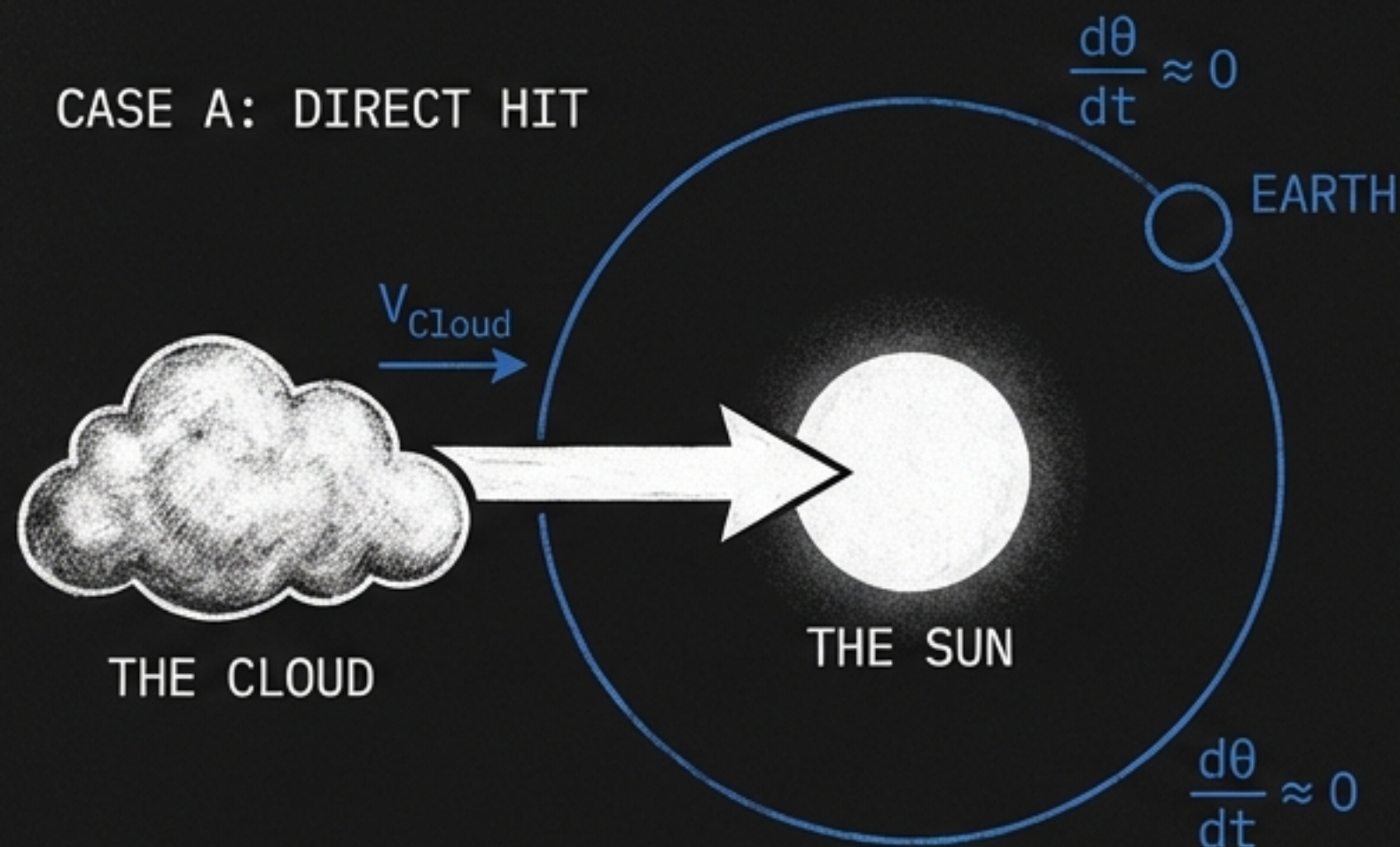
START:
Jan 1964
(Discovery)

----- transit... -----

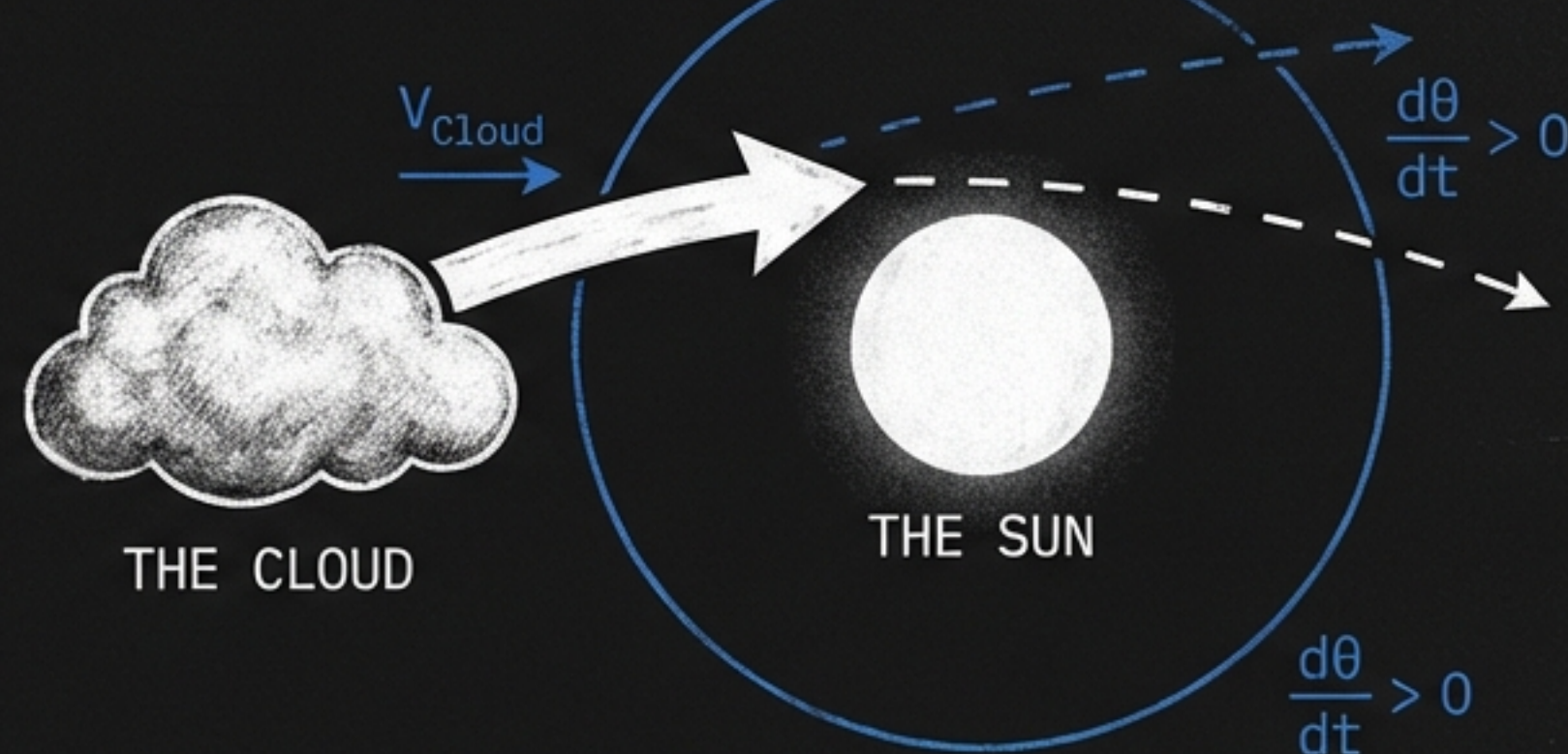
IMPACT:
Aug 1965
(16-18 Months)

DURATION: ~1 Month (Transit time across Earth's orbit)

THE GEOMETRY OF A HIT VS. A MISS



CASE B: NEAR MISS



CRITICAL VARIABLE: Angular Momentum.

SCENARIO A (Direct Hit): Zero angular momentum.
Cloud engulfs Sun. Inner planets immersed.

VERDICT: DARKNESS.

SCENARIO B (Near Miss): Slight lateral velocity. Cloud sweeps past.

EARTH'S POSITION: Earth will be on the far side of the Sun during arrival.

VERDICT: Even if it misses Earth, if it hits the Sun, the light is blocked. Result: DARKNESS.

THERMODYNAMICS OF THE CLOUD

ENERGY EQUATION:

Energy required to lift temperature by T : $1.5 \times 1.3 \times 10^{30} RT$ ergs.

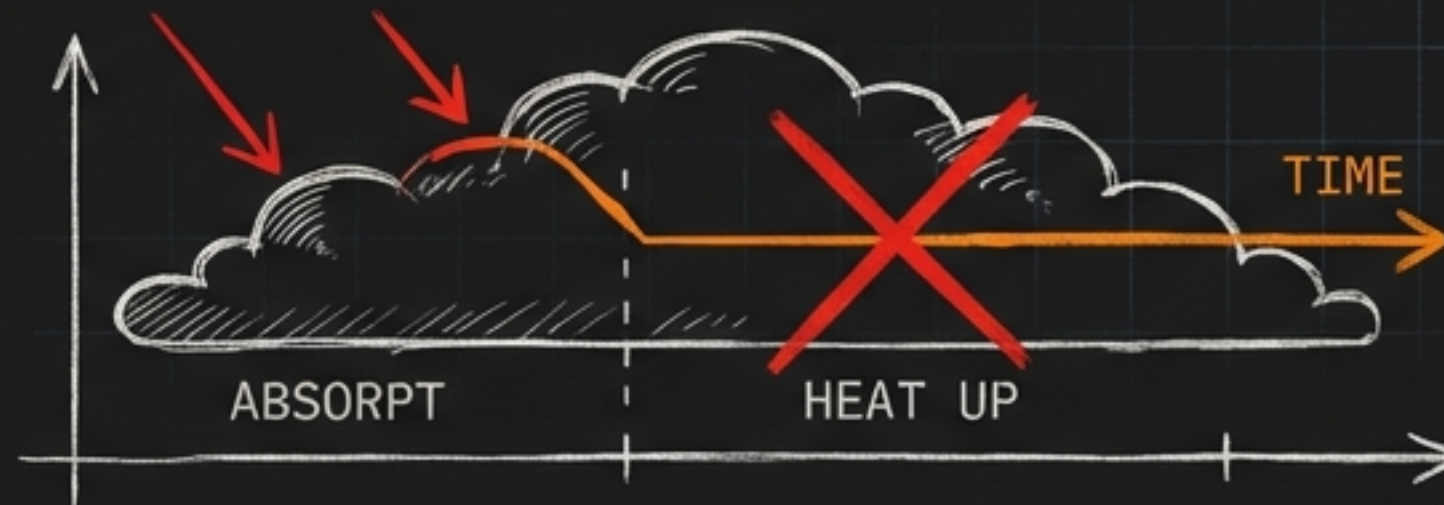
SOLAR OUTPUT (L): 4×10^{33} ergs/sec.

ANALYSIS & CONCLUSION:

CALCULATION: Time to heat cloud to $300K$ = 5 MONTHS of total solar absorption.

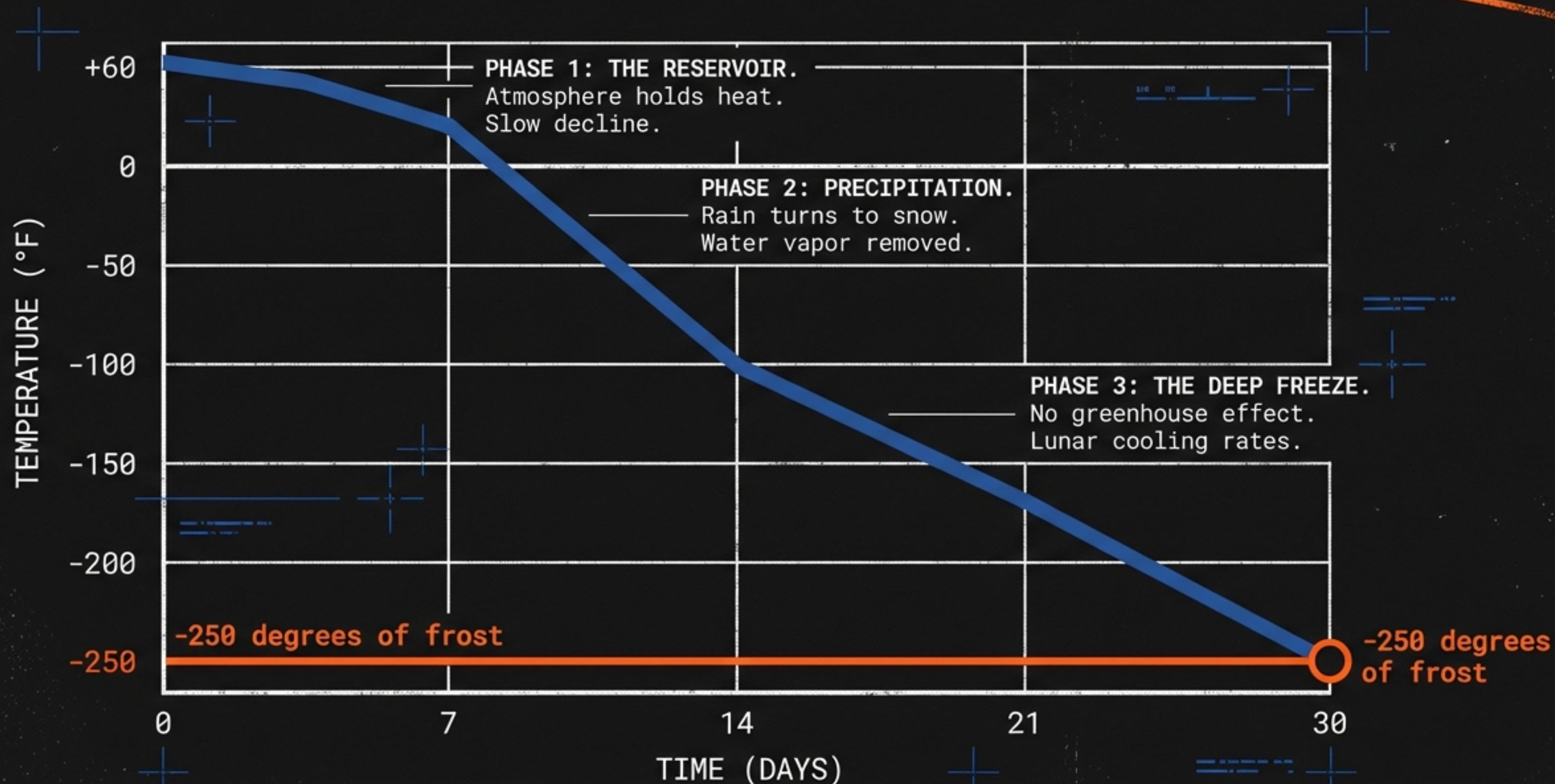
TRANSIT TIME: The Cloud passes in 1 MONTH.

CONCLUSION: The Cloud will not heat up. It will remain cold and opaque.
No radiation will reach Earth.



PROJECTED GLOBAL SURFACE TEMPERATURE

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= EYES ONLY



BIOLOGICAL IMPACT ASSESSMENT

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FLORA:



EXTINCTION.

All growing plants killed.
Seeds likely to survive.

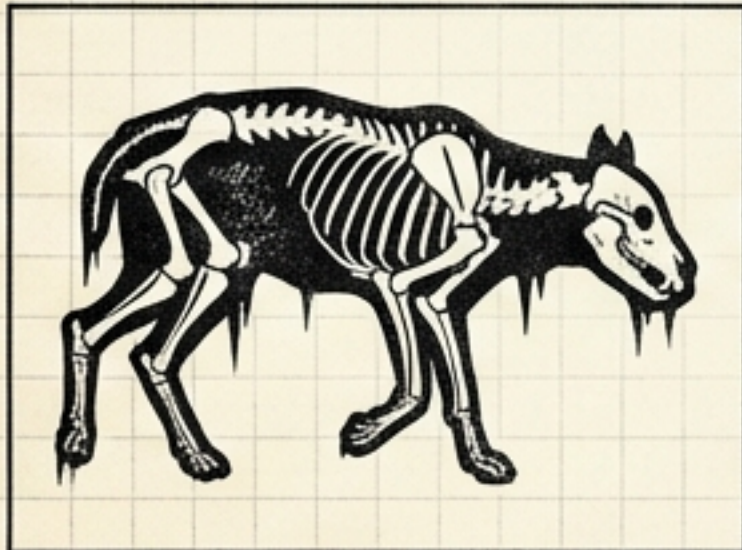
MARINE LIFE:



SAFE.

Oceans act as heat
reservoir. Deep water
remains warm under
surface ice.

LAND ANIMALS:



EXTINCTION.

Total collapse of
surface fauna.

HUMANITY:



UNSURVIVABLE ON SURFACE.

- Standard buildings insufficient.
- Submarines trapped by ice.
- ONLY VIABLE OPTION: Deep caves with artificial heat/air.

THE POLITICAL FILTER

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- EYES ONLY

SCIENTIFIC CYNICISM - KINGSLEY

"Will the politicians pass statutes to stop the Cloud coming? If they can't, let's leave them out of the picture."

OFFICIAL DUTY - HERRICK

"Duty to democratically elected representatives... maintain silence until Governments have made a pronouncement."

ORETECTOR: RULES
AND

DECISION: RESTRICTED.
DIRECTIVE: Absolute Secrecy.
No Public Warning.

HASA ER

THE WASHINGTON STRATEGY

**CONFIDENTIAL
- EYES ONLY**

BRIEFING: The Cabinet Room.

ECONOMIC ASSESSMENT:

- **Duration:** 1 Month.
- **Fuel Stocks:** Moderate but manageable (if rationed).
- **Food Security:** Harvest gathered by July. Crisis hits August.

PRIORITY ACTION: Transfer fuel to strategic underground shelters.



**STOCKPILE
(ROCKIES)**



**STOCKPILE
(APPALACHIANS)**

COLD WAR SCIENTIFIC REALISM

SELECTION CRITERIA: THE NUCLEUS

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EYES ONLY

THE LOGIC: Not everyone can be saved.

THE TARGET LIST: 1. Government Officials 2. Military Command
3. Essential Scientific Personnel 4. Clergy/Royalty

KINGSLEY'S PREDICTION: "When boiled down and distilled, [this] will turn out to mean the political fraternity, field marshals, kings, archbishops."

Technica lines

RESTRICTED
ACCESS

Blueprint Blue

THE REMAINDER: "You and me, Marlowe my boy, will get the icicle treatment."

HASA: ER

CORETECTOR: RULES

AND

COLD WAR SCIENTIFIC REALISM

VARIABLE: THE TEMPERATURE OF THE CLOUD

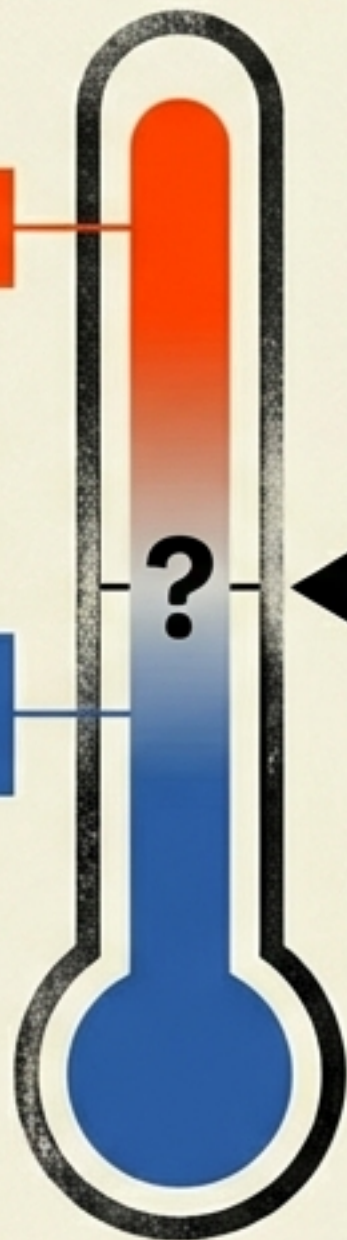
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ZONE 2: HOT CLOUD.

Earth Boils.
Total Biological
Collapse.

ZONE 1: COLD CLOUD.

Earth Freezes.
Survival Possible
in Caves.



THE UNKNOWN: Is the Cloud radiating its own heat?

Blueprint lines

THE RISK: If the Cloud is too hot, even seeds are destroyed.

REQUIRED ACTION: Immediate Radio Astronomy (21cm line) to measure internal temperature.

RESTRICTED
ACCESS

HASA: ER

CORETECTOR: RULES

AND

LIFE CONTINUES IN THE SHADOW

PASADENA:
The Crookshank Estate.

THE CONTRAST:
Real estate tycoons
and film moguls discuss
the future, unaware
there is no future.

RESTRICTED ACCESS

KINGSLEY'S REFLECTION: "It's odd to think that every one of us probably only has a little more than a year to live."

CONFIDENTIAL

HASA: ER

EXECUTIVE SUMMARY: CLASSIFIED

TOP SECRET

Courier Prime

1. **EVENT:** Interstellar gas cloud entry into Solar System.
2. **IMPACT DATE:** July/August 1965.
3. **DURATION:** Approx. 1 Month.
4. **EFFECT:** Total solar obscuration.
5. **PROJECTED TEMP:** -100°F to -250°F .

6. **DIRECTIVES:**

- Maintain absolute secrecy.
- Begin strategic fuel stockpiling.
- Prepare underground shelters for essential personnel.



Blueprint Blue
(#2E5FAC)

RESTRICTED ACCESS

16 MONTHS REMAINING

STATUS: 21.3 AU Distance. Accelerating.

Imperative you and Astronomer Royal
come immediately repeat immediately...
The die is cast.

CONFIDENTIAL
PHN/OTOSTIAL

RESTRICTED
ACCESS